Prototype-Guided Multitask Adversarial Network for Cross-Domain LiDAR Point Clouds Semantic Segmentation

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***Abstract*— Unsupervised domain adaptation (UDA) segmentation aims to leverage labeled source data to make accurate predictions on unlabeled target data. The key is to make the segmentation network learn domain-invariant representations. In this work, we propose a prototype-guided multitask adversarial network (PMAN) to achieve this. First, we propose an intensity-aware segmentation network (IAS-Net) that leverages the private intensity information of target data to substantially facilitate feature learning of the target domain. Second, the category-level cross-domain feature alignment strategy is introduced to flee the side effects of global feature alignment. It employs the prototype (class centroid) and includes two essential operations: 1) build an auxiliary nonparametric classifier to evaluate the semantic alignment degree of each point based on the prediction consistency between the main and auxiliary classifiers and 2) introduce two class-conditional point-to-prototype learning objectives for better alignment. One is to explicitly perform category-level feature alignment in a progressive manner, and the other aims to shape the source feature representation to be discriminative. Extensive experiments reveal that our PMAN outperforms state-of-the-art results on two benchmark datasets.**

***Index Terms*— Adversarial learning, multitask learning, point cloud, semantic segmentation, unsupervised domain adaptation (UDA).**

I. Introduction

SEMANTIC segmentation of 3-D LiDAR point clouds is of great significance to 3-D spatial perception and understanding, including smart cities [1], [2], 3-D map construction [3], autonomous driving [4], [5], etc. Although the performance of supervised segmentation methods [6], [7], [8], [9] has grown immensely, the improved performance is accompanied by laborious manual pointwise annotations. To relieve the heavy burden of human labor, we intend to label the real-world data with the help of synthetic data whose annotations can be obtained at almost zero cost. However, the model trained by synthetic data (source domain) cannot guarantee reliable performance on the real-world data (target domain) due to the different data distributions (domain shift). Thus, researchers have recently explored extensive works called transfer learning or unsupervised domain adaptation (UDA) to address this issue [10].

The fundamental goal of UDA segmentation is to bridge the domain gap between the source and target domains, which learns the domain-invariant features to make the model perform effectively on the target domain. Generally, current UDA segmentation approaches are mainly designed for 2-D images and can be roughly categorized into two branches: adversarial training [11], [12], [13], [14] and self-training [15], [16], [17]. The former learns the domain-invariant feature by playing a two-player min-max game between the generator and discriminator, whereas the latter iteratively retrains the model with confident pseudo-label. Nevertheless, self-training heavily relies on well-initialized parameters and commonly uses adversarial-based method to warm up. For 3-D point clouds, research in both branches is in its infancy, and the previous study [18] shows that the methods particularly designed for 2-D images do not generalize well to point clouds. This motivates us to design an adversarial-based UDA algorithm focused on 3-D point clouds.

As shown in Fig. 1, the sampling patterns mismatch is the major obstacle for 3-D point cloud UDA segmentation task. Despite significant similarities shared by the generator output (e.g., scene layout and the size of the interest objects), this issue still exists. The discriminator can easily make accurate predictions given the different sampling patterns, making it difficult for the generator to learn domain-invariant feature. Therefore, many specifically designed methods [18], [19], [20] are proposed to tackle this issue and made significant progress. However, they align the marginal distributions of the two domains globally, ignoring the alignment difficulty of each point in the training process. One potential adverse effect is that some initially well-aligned points may be incorrectly mapped leading to negative transfer [14]. Furthermore, such global-level alignment will inevitably deteriorate the discriminability of the classifier [21].

In this work, we propose a *prototype-guided multitask adversarial network* (PMAN) for point cloud UDA segmentation. The key idea of PMAN is twofold. First, Fig. 1 illustrates

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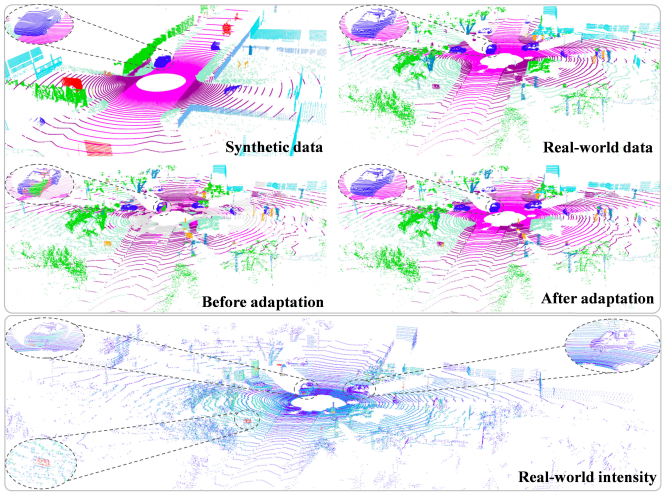

Fig. 1. UDA segmentation for urban scenes 3-D LiDAR point clouds. (Top) Distinct sampling patterns between the synthetic and real-world data, and the real-world segmentation results of our PMAN before and after adaptation. (Bottom) Point intensity visualization. Same class objects have similar intensity (e.g., the traffic sign is high, and the car is low).

that objects of the same class generally have similar intensity (e.g., the traffic sign is high and the car is low), implying that intensity could convey semantic information to a certain extent. In light of this, we propose a multitask network named *intensity-aware segmentation network* (IAS-Net), which incorporates an auxiliary self-supervised intensity regression task to enhance the perception ability of the target domain. Second, considering that the same class points are clustered around their corresponding prototype (class centroid) and geometrically close in the latent feature embedding space, we introduce the fixed prototype into our PMAN to eliminate the side effect of global alignment.

Specifically, we build an auxiliary *nonparametric* classifier based on the distance between the feature and prototypes. To accelerate the alignment of hard-adapt points and prevent negative transfer, we adaptively reweight the adversarial loss according to the prediction consistency of the main and auxiliary classifiers. As no additional learnable parameters are required for this auxiliary classifier, the computational cost is negligible. Furthermore, we introduce two class-conditional point-to-prototype learning objectives to perform category-level feature alignment and improve the discriminability of our IAS-Net. The two losses both aim to reduce the distance between the point embedded feature and its corresponding prototype. Since the target data is unlabeled, the pseudo-label of each target point is employed to minimize the distance between the point feature and its assigned prototype. To prevent the adverse effects of the incorrect pseudo-label, a progressive alignment strategy is introduced. On the other hand, for the richly labeled source data, we use the ground-truth label to enforce that each class feature distribution is structured and compact. The main contributions of this work are summarized as follows.

1) We propose a novel 3-D LiDAR point clouds UDA framework, termed PMAN, which has a multitask segmentation network and performs the category-level alignment.
2) We propose an IAS-Net that utilizes the private intensity information of the target domain to learn generic features and is biased toward learning target domain knowledge.
3) We introduce the prototype into PMAN to eliminate the side effect of global-level alignment and perform category-level alignment in a progressive manner.
4) Extensive experiments on two real-world datasets reveal that the proposed PMAN exceeds the state-of-the-art UDA approaches by a large margin.

II. Related Work

A. Semantic Segmentation for 3-D Point Clouds

Current supervised 3-D semantic segmentation algorithms can be grouped into three categories: point-based, voxel-based, and projection-based. The point-based methods directly utilize multilayer perceptron (MLP) to extract feature representation from the raw irregular point clouds. PointNet [22] is pioneering work in this line of research. Inspired by PointNet, numerous works [6], [23], [24], [25], [26], [27], [28] are then proposed to learn per-point local features. Another line of research is projection-based methods, which project 3-D point clouds onto 2-D images and utilize the mature 2-D image segmentation methods to extract features. RangeNet++ [29], SqueezeSeg [30], SqueezeSegV2 [31], and SalsaNext [4] are examples of these approaches. Despite their high efficiency, they fail to fully exploit the geometric information of 3-D point clouds. Alternatively, we can also discretize point clouds into a 3-D volumetric representation. For example, SparseConvNet [32], MinkUnet [7], RPV-Net [8], and Cylinder3D [9] mainly utilize 3-D sparse convolution to extract per-point features and achieve leading performance. Despite their significant successes on large-scale outdoor datasets, the works mentioned above need tremendous manually annotated training data, which is extremely tedious and time-consuming.

B. UDA for 2-D Images

Over the past decade, UDA technology has been widely explored in 2-D images and has achieved rapid development. The dominant strategy is using adversarial learning to reduce the domain gap. FCNs in the wild [33] is the first work that applies a discriminator operating on the intermediate layers to learn domain-invariant features. Later on, some works [11], [13] suggest that the output space shows more similarities than visual appearance. They perform adversarial learning on the output space and show superior performance over feature space alignment. However, aligning the marginal distribution of the two domains globally would hurt the already mapped region and lead to negative transfer, especially under extreme class imbalance circumstances. Therefore, CLAN [14] and DAST [16] propose to align the feature distribution at the category-level to improve the final performance. FADA [12] adopts a fine-grained domain discriminator, which explicitly incorporates class information into the discriminator. DAST [16] utilizes an attention mechanism that forces the segmentation network to focus on the hard-adapted regions to improve the final segmentation performance. Draw on the experience of 2-D images, for outdoor 3-D LiDAR

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point clouds, it is wiser to reduce the domain gap from category-level.

C. UDA for 3-D Point Clouds

For 2-D images and 3-D point clouds, the causes of domain gaps are different, so the particularly designed 2-D UDA algorithm cannot generalize effectively to 3-D point clouds. Recently, UDA for 3-D point clouds has attracted researcher attention and has been wildly explored. PointDAN [34] is a pioneer work that introduces a self-adaptive node module to align the local features and utilizes adversarial training to reduce the domain gap from the global-level. DefRec [35], GAST [36], Luo et al. [37], and Fan et al. [38] propose to utilize hand-craft or learnable self-supervised tasks to enhance adaptability. DFAN [39] proposes a dual-branch feature alignment network that jointly aligns the global and local features. Shen et al. [40] explicitly models the geometry of the point clouds to help the network to learn geometry-aware information. However, these methods are mainly designed for small-scale datasets.

For urban scenes 3-D LiDAR UDA segmentation tasks, there also emerges some innovative works that have made significant progress. For example, SqueezeSegV2 [31] proposes a multistage training pipeline and a context aggregation module to reduce the sensitivity to dropout noise. Langer et al. [41] proposes to generate a semisynthetic dataset by fusing sequential scans into a dense mesh and resampling them by adopting the target domain LiDAR sensor settings. Complete & Label [18] completes the different domain data into a canonical domain to bridge the domain gap. LiDARNet [19] and Rochan et al. [20] are two multitask adversarial-based UDA methods. They project 3-D point clouds onto 2-D images and employ additional branches/modules to learn domain-specific information. PCT [42] is the latest syn-to-real approach which proposes a translator to minimize the domain gap by changing the appearance and sparsity of the synthetic data. Despite their great success, the performance can be further improved.

III. Method

In cross-domain outdoor point clouds UDA segmentation, we have a labeled source domain $\mathcal{D}^{\mathcal{S}} = \{x^s, y^s\}$ and an unlabeled target domain $\mathcal{D}^{\mathcal{T}} = \{x^t\}$, where x is a scan and y is the corresponding point-level annotation. $\mathcal{D}^{\mathcal{S}}$ and $\mathcal{D}^{\mathcal{T}}$ share the same label space, but are with different joint distributions $P(x^s, y^s) \neq P(x^t, y^t)$. We aim to use both the labeled $\mathcal{D}^{\mathcal{S}}$ and unlabeled $\mathcal{D}^{\mathcal{T}}$ to learn a model that obtains accurate predictions on $\mathcal{D}^{\mathcal{T}}$. To achieve this, we propose PMAN, which comprises a segmentation network G (i.e., IAS-Net) and a discriminator D. In the following, we first provide the detailed network architecture of IAS-Net. Then, we present how to utilize the prototype to perform category-level alignment and enhance the discriminability of IAS-Net in PMAN framework.

A. Intensity-Aware Segmentation Network

The data collected by a LiDAR sensor typically contains four attributes, XYZ coordinates and reflection intensity. The intensity is the return strength of a laser beam which is affected by numerous factors. The two most significant of which are the scanning angle and the surface composition of the scanned object. Typically, objects in the same class commonly have similar surface compositions, e.g., the building surface is concrete, and the car is made of metal. In light of this, the proposed IAS-Net uses the private intensity information of the target data to increase the perception ability of the target domain knowledge.

As shown in Fig. 2, the IAS-Net is a multitask network with a shared encoder (Enc), two-branch decoders (Dec_s and Dec_r) and two classifiers (C_m and C_a). The backbone network is the standard MinkUnet14A [7] which gives each input point a semantic prediction. The Enc and Dec_s are with four downsample and upsample layers, the Dec_r is with two upsample layers, and C_m is a standard 3-D sparse convolution layer. C_a is a nonparametric classifier, and we will present the details in Section III-C. After the Dec_s performs two upsampling operations, we graft the Dec_r onto the Dec_s to conduct the auxiliary intensity regression task for target domain data. Thus, the Dec_r will facilitate IAS-Net to learn the target domain features. Specifically, the Dec_r only consists of two upsampling layers, where the channel number is {64, 32}. Each upsampling layer consists of one sparse transpose convolution, standard sparse convolution, batch normalization, and Leaky-Relu layer. The regressor R_{avg} is an average pooling layer that enables the IAS-Net to concentrate on learning relevant information for the primary task without any extra learnable parameters. Benefiting from this simple and practical design of the Dec_r, it can be easily integrated into the existing network with negligible computational cost.

Given a labeled source scan x_i^s with N input points, the IAS-Net is optimized by standard cross-entropy loss to obtain a precise semantic prediction of each input point

$$\mathcal{L}_{\text{seg}}^s = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_i^s \log P_{i,k}^s \tag{1}$$

where K is the number of classes, y_i^s and $P_{i,k}^s$ are the ground-truth semantic label and the probability predictions of the ith point, respectively.

For the unlabeled target domain data x_i^t, we aim to utilize the auxiliary intensity regression task to strengthen the representation power of the IAS-Net. As the Huber loss [43] is robust and generally not sensitive to outliers, we adopt it to optimize this auxiliary task. The loss function is defined as

$$\mathcal{L}_{\text{reg}}^t = \frac{1}{N}\sum_{i=1}^{N} \text{Huber}\left(I_i^t - \hat{I}_i^t\right) \tag{2}$$

$$\text{Huber}(z) = \begin{cases} |z|_1, & \text{if } |z| \leq c \\ \dfrac{z^2 + c^2}{2c}, & \text{otherwise} \end{cases} \tag{3}$$

where $c = 0.1$ by default. I_i^t and $\hat{I}_i^t$ are the output of R_{avg} and ground-truth intensity value, respectively.

Notably, the Dec_r is *only* trained by the target domain data. We do not fuse the feature of the Dec_r into the Dec_s, and the Dec_r can be removed at inference time. This design allows

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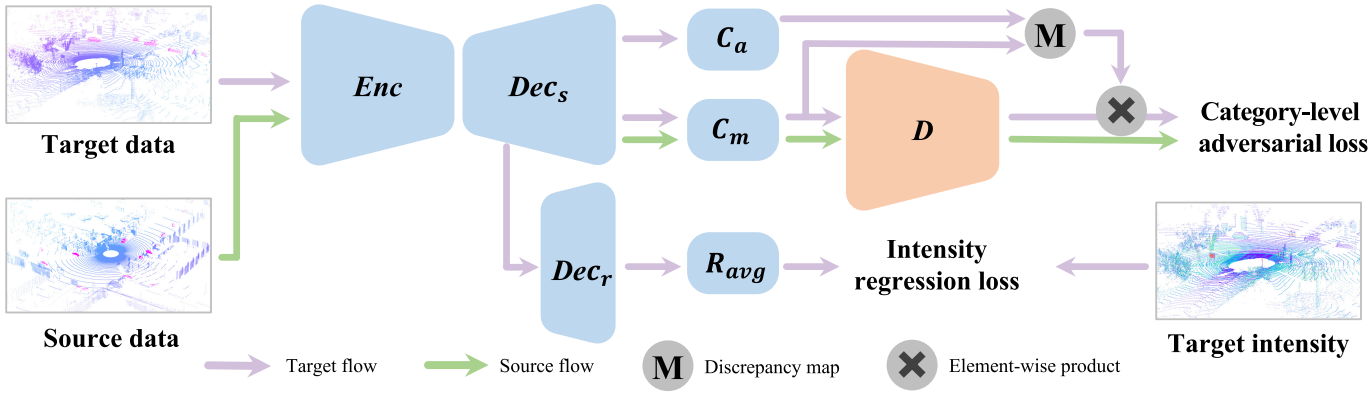

Fig. 2. Overview of our PMAN with a multitask segmentation network IAS-Net and a discriminator D. The IAS-Net has a shared encoder Enc and two separated decoders. Dec_s and Dec_r perform the primary segmentation and auxiliary intensity regression tasks, respectively. Besides the regressor R_{avg}, there exists two classifiers C_m and C_a. The *nonparametric classifier* C_a determines the class of each point based on the distance between the embedded feature and prototypes. Furthermore, the prediction consistency between C_m and C_a is used to generate a discrepancy map to weight the adversarial loss adaptively. Segmentation loss is omitted in the figure to simplify the illustration.

the self-training UDA segmentation method directly use our method for warm-up without any additional processing (e.g., model distillation). Furthermore, the Dec_r is independent of the segmentation network architecture and can be grafted onto any backbone segmentation network.

B. Prototype-Guided Adversarial Alignment

As the target data is unlabeled, traditional adversarial approach [13] adopts the adversarial loss $\mathcal{L}_{\text{adv}}^t$ to align the two domain marginal distribution at global-level

$$S_i^t = -P_i^t \log P_i^t \tag{4}$$

$$\mathcal{L}_{\text{adv}}^t = -\frac{1}{N}\sum_{i=1}^{N} \left|\left|D\left(S_i^t\right) - 0\right|\right|_2 \tag{5}$$

where 0 and S_i^t, respectively, denote the source domain label and self-information [13]. $\mathcal{L}_{\text{adv}}^t$ is the least square loss [44]. However, this global-level alignment ignores the underlying structures information among classes. The extracted feature of the target domain data is forced to match the source domain feature distribution, and some points are easily incorrectly mapped. Following the spirit of Cotraining [45] and CLAN [14], we employ two separate classifiers with different views to tackle this problem without introducing any learnable parameters.

For a well-trained discriminative model, the same class points should cluster around their corresponding prototype (class centroid) in the latent feature space. Inspired by CAG-UDA [15], we construct an auxiliary *nonparametric classifier* C_a to predict the probability that a point x_i^t belongs to class k at the feature space. Formally, the probability is calculated by the distance between the embedded feature $\text{Dec}_s(\text{Enc}(x_i^t))$ and the prototype λ_k^s

$$p\left(y_i^t = k | x_i^t\right) = \frac{\exp\left(-d\left(\text{Dec}_s\left(\text{Enc}\left(x_i^t\right)\right), \lambda_k^s\right)\right)}{\sum_k \exp\left(-d\left(\text{Dec}_s\left(\text{Enc}\left(x_i^t\right)\right), \lambda_k^s\right)\right)} \tag{6}$$

where d is a distance metric (e.g., L2 distance as in [15]), λ_k^s is the prototype of class k in the source domain

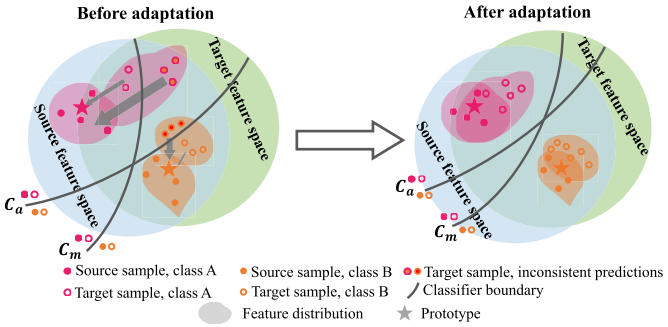

Fig. 3. Illustration of the self-adaptive adversarial alignment. Two classifiers C_a and C_m are employed to give each point a semantic prediction from two distinct views. The self-adaptive weight of each point is calculated based on the consistency of the two view predictions. The point with inconsistent predictions will be given a large weight (gray arrow) to accelerate the alignment, and vice versa. The thicker the gray arrow, the higher the weight.

and calculated by

$$\lambda_k^s = \frac{1}{\left|\mathcal{D}_k^S\right|} \sum_{x_i^s \in \mathcal{D}_k^S} \text{Dec}_s\left(\text{Enc}\left(x_i^s\right)\right) \tag{7}$$

where $\mathcal{D}_k^S$ is the subset of $\mathcal{D}^S$, including all the points whose labels are k in the source domain. Notably, because of the limited batch size and extreme class imbalance of source data distribution, we *fixed* the prototype during training.

As shown in Fig. 3, we evaluate how well each point is semantically matched based on the prediction consistency of C_m and C_a. Specifically, since C_m is trained by the annotated source domain data with cross-entropy loss, it is biased toward the source domain and suffers from the class imbalance problem. In contrast, C_a is not dependent on the biased classifier and treats each class equally [15]. It can give each point a reliable prediction distinct from the view of C_m. When two classifiers give consistent predictions for the same point, the point is semantically aligned to a large extent. In this case, we give the adversarial loss a tiny weight (i.e., the thinner gray arrow in Fig. 3) to prevent negative transfer. In contrast, inconsistent predictions (i.e., the point with different colors in Fig. 3) denote this point is near the decision boundary and is far from being correctly mapped, and a large weight (i.e., the

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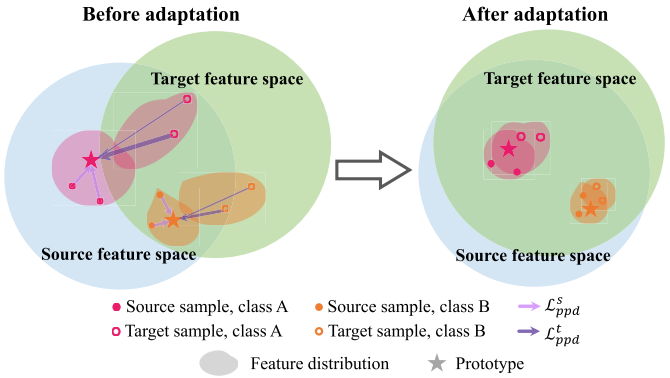

Fig. 4. Illustration of the two point-to-prototype losses. $\mathcal{L}^t_{\text{ppd}}$ aims to align the identical category feature distributions in two domains progressively. The thicker the arrow, the higher the weight. $\mathcal{L}^s_{\text{ppd}}$ is intended to shape the source data feature representation into discriminative clusters.

thicker gray arrow in Fig. 3) should be given to the adversarial loss to accelerate the alignment. To adaptively reweighted the adversarial loss, we use the cosine similarity to generate a discrepancy map $\mathcal{M} \in \mathbb{R}^{N\times 1}$ from the C_m and C_a outputs

$$\mathcal{M}(P_m, P_a) = 1 - \frac{\langle P_m, P_a\rangle}{\|P_m\| \cdot \|P_a\|} \tag{8}$$

where P_m and P_a are the predictions of C_m and C_a, respectively.

The self-adaptive adversarial loss $\mathcal{L}^{t,\text{ada}}_{\text{adv}}$ is

$$\mathcal{L}^{t,\text{ada}}_{\text{adv}} = -\frac{1}{N}\sum_{i=1}^{N}(\gamma_{\text{local}}\mathcal{M} + \epsilon)\left|\left|D\left(S^t_i\right) - 0\right|\right|_2 \tag{9}$$

where γ_{local} and ϵ are, respectively, controlling the self-adaptive and global adversarial losses.

Compared with previous methods [11], [13] which bridge the domain gap from the global-level, we incorporate semantic consistency into this self-adaptive adversarial loss and term it category-level adversarial alignment. Besides, our approach is without the requirements of extra learnable parameters and loss functions, which is much simpler than CLAN [14].

C. Progressive Category-Level Feature Alignment

To generate better alignment and further improve the final segmentation performance, as shown in Fig. 4, we utilize the prototype to *progressively* align the categorywise feature distribution across domains. Since the label is unavailable for the target data, we employ a confident strategy to obtain the pseudo-label, i.e., target points with both larger prediction probability $p^t_i = [p^t_{i,1}, \ldots, p^t_{i,K}]$ and lower entropy e^t_i have better prediction accuracy, where e^t_i is calculated by

$$e^t_i = -\frac{1}{\log K}\sum_{k=1}^{K} p^t_{i,k}\log\left(p^t_{i,k}\right). \tag{10}$$

We set two confidence thresholds Th_p and Th_e for p^t_i and e^t_i to obtain the pseudo-label, respectively. Formally, the pseudo-label $\hat{y}^t_i$ of the ith point can be obtained by

$$\hat{y}^t_i = \begin{cases} \arg\max_k p^t_i, & \max\left(p^t_i\right) > \text{Th}_p \text{ and } e^t_i < \text{Th}_e \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

With the help of the pseudo-label, we utilize the point-to-prototype distance loss $\mathcal{L}^t_{\text{ppd}}$ to explicitly align the feature distributions at category-level

$$\mathcal{L}^t_{\text{ppd}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\mathbb{I}\left(\hat{y}^t_i = k\right)\left|\left|\text{Dec}_s\left(\text{Enc}\left(x^t_i\right)\right) - \lambda^s_k\right|\right|_1. \tag{12}$$

Even under such strict constraints, we cannot guarantee that the pseudo-labels must be correct, and some inaccurate pseudo-labels could still exist. The inaccurate pseudo-labels will cause error accumulation and compromise the subsequent learning. To remedy this problem, we propose a progressive alignment strategy that minimizes the $\mathcal{L}^t_{\text{ppd}}$ with self-adaptive weights. Specifically, we measure the cosine similarity $d^t_{\cos}(x^t_i)$ between the embedded feature $\text{Dec}_s(\text{Enc}(x^t_i))$ and the corresponding prototype $\lambda^s_{k=\hat{y}^t_i}$ of each target point x^t_i. The advantage of cosine similarity is that its value range is fixed, i.e., $[-1, 1]$. To conveniently train, $d^t_{\cos}$ is further normalized to $[0, 1]$

$$d^t_{\cos}\left(x^t_i\right) = 0.5 \cdot \frac{\left\langle \text{Dec}_s\left(\text{Enc}\left(x^t_i\right)\right), \lambda^s_{k=\hat{y}^t_i}\right\rangle}{\left\|\text{Dec}_s\left(\text{Enc}\left(x^t_i\right)\right)\right\| \cdot \left\|\lambda^s_{k=\hat{y}^t_i}\right\|} + 0.5. \tag{13}$$

The small $d^t_{\cos}(x^t_i)$ indicates the distance between $\text{Dec}_s(\text{Enc}(x^t_i))$ and $\lambda^s_{k=\hat{y}^t_i}$ is large (i.e., closer to the decision boundary and farther from the assigned prototype) in the feature space. In this case, we should give $\mathcal{L}^t_{i,\text{ppd}}$ a small weight to prevent $\hat{y}^t_i$ is an incorrect pseudo-label with high confidence. In contrast, the more similar of $\text{Dec}_s(\text{Enc}(x^t_i)))$ and $\lambda^s_{\hat{y}^t_i}$, the larger the weight. With the decrease of the domain gap in the subsequent training, the similarity $d^t_{\cos}$ would increase progressively. In implementation, we adopt an elementwise product to apply this self-adaptive weight $d^t_{\cos}$ to $\mathcal{L}^t_{\text{ppd}}$.

Moreover, the adversarial-based UDA method inevitably deteriorates the discriminability of G and results in performance degradation [21]. Although the cross-entropy loss can partially ensure the intraclass and interclass distances are compactness and dispersion, respectively, it only optimizes the relative relations between intraclass and interclass distances [46]. To further increase the discriminability of IAS-Net, we use the prototype to shape the source feature representation into discriminative clusters. Since the source data are precisely labeled, we leverage the $\mathcal{L}^s_{\text{ppd}}$ to explicitly minimize the distance between the embedded feature $\text{Dec}_s(\text{Enc}(x^s_i))$ and prototype $\lambda^s_{y^s_i}$

$$\mathcal{L}^s_{\text{ppd}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\mathbb{I}\left(y^s_i = k\right)\left|\left|\text{Dec}_s\left(\text{Enc}\left(x^s_i\right)\right) - \lambda^s_k\right|\right|_1. \tag{14}$$

In summary, PMAN is trained by the following total loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}^s_{\text{seg}} + \gamma_1\mathcal{L}^t_{\text{reg}} + \gamma_2\mathcal{L}^{t,\text{ada}}_{\text{adv}} + \gamma_3\left(\mathcal{L}^s_{\text{ppd}} + \mathcal{L}^t_{\text{ppd}}\right) \tag{15}$$

where γ_1, γ_2, and γ_3 denote hyperparameters that, respectively, trade-off the contribution of each loss.

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IV. Experiments and Results

A. Datasets and Class Mapping

We set the synthetic dataset as the source domain, and validate the effectiveness of our PMAN with two real-world datasets collected by distinct LiDAR at various places.

SynLiDAR [42] is a recently proposed large-scale synthetic dataset with 13 sequences (19 840 scans) and 32 semantic classes. It contains a variety of realistic virtual scenes, including cities, towns, harbors, etc. Following the official recommendation, we use all sequences for training.

SemanticKITTI [47] is the wildly used real-world urban scene dataset for evaluating the segmentation algorithms, which is acquired by a Velodyne HDL-64E LiDAR in Germany. The points are accurately labeled into 28 semantic categories. Following the widely used training protocol, we use sequences 00–07, 09–10 for training (19 130 scans), and sequence 08 for testing (3071 scans).

SemanticPOSS [48] is a relatively smaller real-world dataset, which consists of six sequences (2988 scans) and 14 semantic classes. This dataset is acquired by a Pandora 40-line LiDAR on the Peking University. We use sequence 03 (500 scans) for testing and the others (2488 scans) for training.

Following the official recommendation [42], we remap the semantic categories to ensure that class definitions across the source and target domain are consistent. To pair with semanticKITTI and semanticPOSS for UDA segmentation, we map SynLiDAR labels into 19 and 13 semantic classes, respectively. The class mapping details are available at https://github.com/xiaoaoran/SynLiDAR.

B. Implementation Details

All experiments are conducted on a single Tesla V100 GPU and implemented with PyTorch framework and MinkowskiEngine [7]. Both our IAS-Net and the discriminator are voxel-based networks, and the voxel size is 0.05 m. The main segmentation branch of IAS-Net is the standard MinkUnet14A. Like AdaptSegNet [11], the discriminator consists of five sparse convolution layers with kernel size 4 and stride 2, where the channel number is {32, 64, 64, 128, 1}. Each convolution layer, except the last layer, is followed by a Leaky-ReLU parameterized by 0.2. The discriminator is trained without any normalization layers and the output of the discriminator is upsampled using Minkowski interpolation to the input size. Our IAS-Net and discriminator are trained by the Adam optimizer with learning rates of 0.00025 and 0.0001, respectively. The learning rate is decaying by a poly learning rate policy, where the current learning rate equals the initial one multiplying $(1 - (\text{iter}/(\text{max_iter})))^{\text{power}}$ with power $= 0.9$. The maximum number of training iterations is 100 K. In our best model, the hyperparameters γ_1, γ_2, γ_3, γ_{local}, ϵ, Th_p, and Th_e are set to 0.1, 0.001, 1, 80, 0.4, 0.95, and 0.01, respectively.

C. Experimental Results

In this section, we present the experimental results of our PMAN in comparison with various prevalent output space adversarial-based UDA segmentation methods, including AdaptSegNet [11], ADVENT [13], FADA [12], CLAN [14], DAST [16], and MRNet [49]. In addition, we also compare PMAN with three recently proposed self-training methods (i.e., PCT [42], PolarMix [51] and CoSMix [50]) and two conventional supervised point cloud segmentation algorithms (i.e., RandLA-Net [6] and MinkUnet14A [7]) to increase the credibility of PMAN. Specifically, AdaptSegNet is the first adversarial-based output space domain alignment UDA segmentation method. AdvEnt takes a step further, which is the first entropy-based adversarial training approach. Since PMAN is a category-level adversarial alignment method, we also choose three recently proposed 2-D image category-level adversarial-based approaches for comparison (i.e., CLAN, FADA, and DAST). All compared 2-D methods have published their codes. By replacing the 2-D feature extractor G and discriminator D with the 3-D voxel-based feature extractor MinkUnet14A and discriminator, we reimplement these 2-D approaches to make them appropriate for 3-D UDA segmentation tasks. We also searched for the optimum hyperparameters for them. Notably, the self-training part of DAST and MRNet is excluded because all compared adversarial-based methods do not use self-training to boost the final performance. Besides, PCT, PolarMix, and CoSMix are the latest self-training-based works which focusing on 3-D point clouds UDA synthetic-to-real segmentation task. Self-training-based method usually has higher performance than adversarial-based method. While the experiment results shows that PMAN outperforms them by a large margin. Mean Intersection-over-Union (mIoU) is the standard metric for segmentation comparison over all classes.

Furthermore, the intensity is an essential attribute of LiDAR point clouds and is challenging to simulate [42]. Thus, the pointwise intensity value of SynLiDAR dataset is obtained by training *a rendering model from a real-world dataset (i.e., semanticKITTI)*. In Fig. 5, we visualize the intensity of the three datasets (i.e., SynLiDAR, semanticKITTI, and semanticPOSS) utilized in this article. Obviously, the intensity of the SynLiDAR is especially different from that of the semanticPOSS and is similar to the semanticKITTI to a certain extent. To explore the effect of intensity as an input feature on the final performance, we conduct experiments with XYZ coordinates (XYZ) and both XYZ and intensity (XYZI) as an input feature on the SynLiDAR $\rightarrow$ SemanticKITTI and SynLiDAR $\rightarrow$ SemanticPOSS tasks, respectively.

1) SynLiDAR $\rightarrow$ SemanticKITTI With XYZ as Input: As shown in Table I, our PMAN outperforms the other comparison methods by a large margin, achieving an mIoU score of 33.7%. Compared with the source only model, PMAN improves the mIoU from 20.4% to 33.7%, outperforming in all categories and demonstrating that PMAN effectively prevents the negative transfer. PMAN outperforms the second-best GAN-based methods CLAN and AdvEnt 3.2%. In addition, for small objects such as motorcycle, person, and bicyclist, PMAN also gives more accurate predictions than previous methods.

2) SynLiDAR $\rightarrow$ SemanticKITTI With XYZI as Input: Table II reports the final segmentation results with XYZI as input. PMAN achieving an mIoU score of 36.3% and outperforms the source only model by +14.6%. For the source only model, XYZI input improves the mIoU by 1.3%

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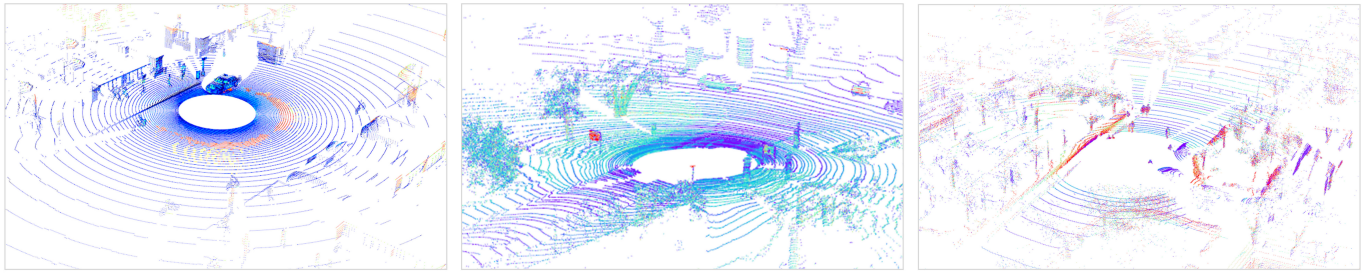

Fig. 5. Intensity visualization of each dataset. Red regions in the each image correspond to high intensity, while blue ones correspond to low intensity.

TABLE I

COMPARISON RESULTS OF SYNLIDAR → SEMANTICKITTI ADAPTATION IN TERMS OF MIOU. THE INPUT FEATURE OF ALL METHODS IS *XYZ* COORDINATES. "SOURCE ONLY" AND "TARGET ONLY" DENOTE THE MODEL ONLY TRAINED ON SOURCE AND TARGET DATA WITHOUT ADAPTATION, RESPECTIVELY. THE MECHANISM "A" AND "S" DENOTE ADVERSARIAL TRAINING AND SELF-TRAINING, RESPECTIVELY

Methods	Mech.	car	bi.cle	mt.cle	truck	oth-v.	pers.	bi.clst	mt.clst	road	parki.	sidew.	other-g.	build.	fence	veget.	trunk	terr.	pole	traf.	mIoU
Source only	-	35.9	7.5	10.7	0.6	2.9	13.3	44.7	3.4	21.8	6.9	29.6	0.0	34.1	7.4	62.9	26.0	35.5	30.3	14.1	20.4
AdaptSegNet [11]	A	52.1	10.8	11.2	2.6	9.6	15.1	35.9	2.6	62.2	10.4	41.3	0.1	58.1	17.1	68.0	38.4	38.7	35.9	20.4	27.9
CLAN [14]	A	51.0	**15.8**	16.8	2.2	7.8	18.7	46.8	3.0	68.9	11.1	44.9	0.1	59.6	17.5	71.7	41.1	44.0	37.7	19.8	30.5
AdvEnt [13]	A	59.9	13.8	14.6	**3.0**	8.0	17.7	45.8	3.0	67.6	**11.3**	45.6	0.1	**61.7**	15.8	**72.4**	41.5	47.0	34.5	15.3	30.5
DAST [16]	A	50.6	9.7	8.2	2.6	6.9	13.1	34.4	2.2	65.2	11.0	41.7	0.1	58.0	14.1	58.6	34.4	35.6	32.7	18.8	26.2
FADA [12]	A	49.9	6.7	5.1	2.5	10.0	5.7	26.6	2.3	65.8	10.8	37.8	0.1	60.3	**21.5**	60.4	37.2	31.9	35.4	17.4	25.6
MRNet [49]	A	49.5	11.0	12.2	2.2	8.6	16.0	46.4	2.7	60.0	10.5	41.9	0.1	55.1	16.5	68.1	38.0	40.7	36.5	**20.8**	28.3
CoSMix [50]	S	56.4	10.2	20.8	2.1	**13.0**	**25.6**	41.3	2.2	67.4	8.2	43.4	0	57.9	12.2	68.4	**44.8**	35.0	**42.1**	17.0	29.9
PMAN (Ours)	A	**71.0**	14.9	**24.8**	1.6	6.6	23.6	**61.1**	**5.5**	**75.3**	10.5	**54.1**	0.1	47.9	17.4	69.6	38.6	**61.5**	37.0	18.6	**33.7**
Target only R [6]	-	91.0	17.1	32.2	42.8	34.2	52.6	69.2	0	90.3	38.6	76.4	3.1	87.2	43.5	84.5	60.1	75.5	50.7	36.0	51.8
Target only M [7]	-	96.2	17.6	55.9	79.5	51.4	65.5	84.9	2.8	93.2	38.5	79.8	1.9	90.4	57.3	86.8	65.4	72.7	64.3	43.2	60.4

TABLE II

COMPARISON RESULTS OF SYNLIDAR → SEMANTICKITTI ADAPTATION IN TERMS OF MIOU. THE INPUT FEATURE OF ALL METHODS IS THE COMBINATION OF *XYZ* COORDINATES AND INTENSITY. "SOURCE ONLY" AND "TARGET ONLY" DENOTE THE MODEL ONLY TRAINED ON SOURCE DATA AND TARGET DATA WITHOUT ADAPTATION, RESPECTIVELY. THE MECHANISM "A" AND "S" DENOTE ADVERSARIAL TRAINING AND SELF-TRAINING, RESPECTIVELY

Methods	Mech.	car	bi.cle	mt.cle	truck	oth-v.	pers.	bi.clst	mt.clst	road	parki.	sidew.	other-g.	build.	fence	veget.	trunk	terr.	pole	traf.	mIoU
Source only	-	42.5	8.6	14.6	0.6	3.8	15.3	49.0	1.3	19.4	5.5	28.0	0.1	37.8	10.3	63.1	31.5	41.6	28.1	11.3	21.7
AdaptSegNet [11]	A	61.5	15.8	15.4	2.1	7.3	17.1	45.9	2.5	63.0	10.2	43.2	0.1	55.4	14.2	67.2	39.4	42.6	31.9	18.9	29.1
CLAN [14]	A	61.9	**20.1**	27.2	2.5	**9.6**	20.1	55.2	**4.1**	68.9	11.5	46.2	0.1	60.9	16.2	72.8	41.8	42.0	34.9	18.9	32.4
AdvEnt [13]	A	63.5	19.0	18.1	**2.8**	8.7	17.3	48.4	2.2	72.1	11.6	47.2	0.1	60.0	17.8	71.5	41.8	47.7	36.0	**20.0**	31.9
DAST [16]	A	60.8	16.7	19.5	2.1	7.2	17.5	39.6	2.4	65.2	11.1	43.7	0.1	57.0	16.4	56.6	38.8	35.5	31.1	17.7	28.4
FADA [12]	A	61.4	14.0	12.7	1.1	1.6	17.6	31.9	1.9	64.6	10.4	38.5	0.1	59.5	15.8	56.6	39.3	39.3	33.8	19.6	27.3
MRNet [49]	A	62.2	16.1	16.7	2.4	7.8	17.9	41.6	2.1	65.7	10.8	43.9	0.1	55.4	15.4	66.6	40.9	40.5	31.7	17.8	29.2
PCT [42]	S	70.8	7.3	13.1	1.9	8.4	12.6	44.0	0.6	56.4	4.5	31.8	0.0	**66.7**	**23.7**	73.3	34.6	48.4	39.4	11.7	28.9
PolarMix [51]	S	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	31.0
CoSMix [50]	S	61.4	8.6	24.0	0.9	5.3	25.6	**55.5**	3.6	58.0	8.3	41.8	0	58.7	16.4	73.5	41.5	36.7	**44.7**	14.4	30.5
PMAN (Ours)	A	**74.9**	18.0	**33.3**	1.9	7.8	**28.7**	52.7	4.0	**77.6**	**12.6**	**56.9**	0.1	63.8	17.9	**74.1**	**43.8**	**63.5**	38.3	19.3	**36.3**
Target only R [6]	-	91.1	35.8	41.9	41.7	32.9	53.7	69.0	0	92.0	38.5	78.1	0.7	85.8	42.7	84.8	57.9	74.4	51.1	40.5	53.3
Target only M [7]	-	95.4	42.7	62.6	82.1	53.1	70.7	87.6	0	93.7	41.5	81.2	3.0	89.4	56.3	88.0	65.4	75.0	65.0	50.6	63.3

compared with *XYZ* input. Compared with PCT, specifically designed for 3-D UDA segmentation, we obtain +7.4% mIoU improvement and give more accurate predictions for all small objects except the pole class. PMAN also outperforms PolarMix and CoSMix, the two latest self-training-based methods, 5.3% and 5.8%, respectively. Moreover, since the intensity values of the SynLiDAR dataset are rendered by a model trained on the semanticKITTI dataset, the intensity values of the two datasets show extremely high similarities. Using intensity as an additional input feature can boost the final performance. The experimental result of all compared approaches reported in Tables I and II demonstrate that the intensity information *benefits* the final segmentation performance in this cross-domain UDA segmentation

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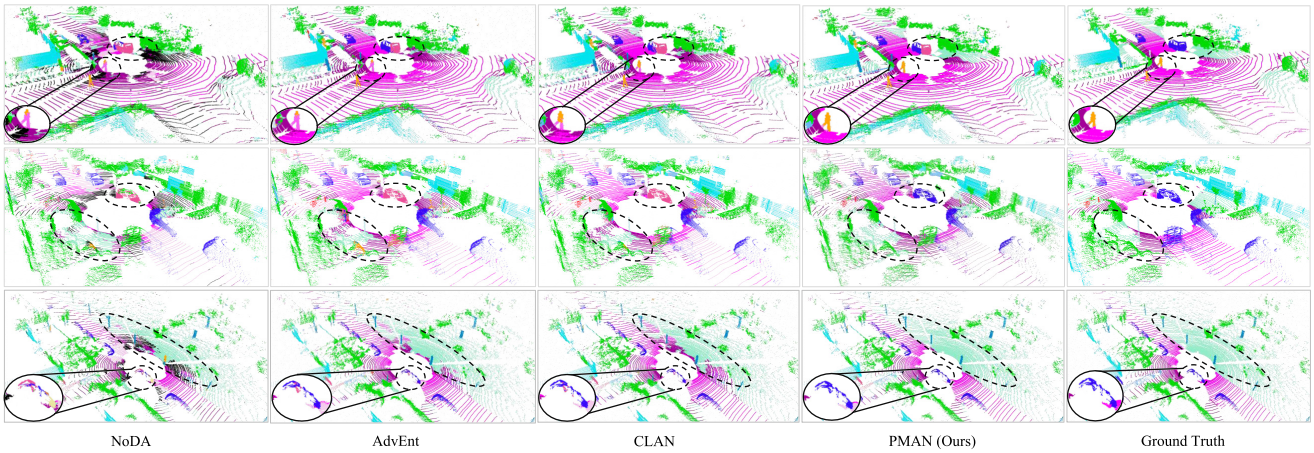

Fig. 6. Visual results of UDA segmentation for SynLiDAR → SemanticKITTI task. Black circles highlight some regions of interest.

TABLE III

COMPARISON RESULTS OF SYNLIDAR → SEMANTICPOSS ADAPTATION IN TERMS OF MIOU. THE INPUT FEATURE OF ALL METHODS IS *XYZ* COORDINATES. "SOURCE ONLY" AND "TARGET ONLY" DENOTE THE MODEL ONLY TRAINED ON SOURCE DATA AND TARGET DATA WITHOUT ADAPTATION, RESPECTIVELY. THE MECHANISM "A" AND "S" DENOTE ADVERSARIAL TRAINING AND SELF-TRAINING, RESPECTIVELY

Methods	Mech.	bi.clst	car	trunk	veget.	traf.	pole	garb.	build.	cone.	fence	bi.cle	ground	pers.	mIoU
Source only	-	47.2	43.6	37.8	70.3	11.1	33.8	19.5	67.9	11.2	19.9	9.6	77.9	47.8	38.3
AdaptSegNet [11]	A	43.9	48.2	39.0	69.6	15.5	33.6	21.3	64.3	12.7	25.0	11.6	76.0	49.9	39.3
CLAN [14]	A	43.9	46.6	41.3	71.0	15.1	34.3	20.4	69.6	9.5	23.2	12.0	75.1	51.3	39.5
AdvEnt [13]	A	44.6	47.6	40.3	71.2	15.6	35.6	22.0	68.4	10.6	25.9	10.4	76.7	52.3	40.1
DAST [16]	A	44.6	46.1	39.7	71.1	17.7	34.0	20.3	68.7	11.4	23.0	11.6	76.2	50.3	39.6
FADA [12]	A	39.6	41.2	38.8	69.2	16.3	32.1	18.1	67.9	11.5	22.0	13.0	71.4	47.9	37.6
MRNet [49]	A	43.5	47.2	39.1	70.4	15.5	32.8	22.0	66.1	13.2	24.2	11.2	76.8	50.0	39.4
CoSMix [50]	S	**53.6**	47.6	44.8	**75.1**	16.8	**37.9**	**25.3**	72.7	**19.9**	**39.7**	10.8	**80.0**	56.5	44.6
PMAN (Ours)	A	52.6	**61.5**	**46.8**	**75.1**	**18.8**	36.5	21.4	**74.7**	18.3	25.8	**37.5**	73.7	**61.9**	**46.5**
Target only R [6]	-	48.2	50.7	46.4	73.9	37.5	24.0	9.1	76.3	39.9	46.7	49.4	79.0	50.5	48.6
Target only M [7]	-	61.6	75.1	48.4	77.9	47.7	37.8	29.9	77.8	37.7	51.2	54.9	81.2	63.9	57.3

task. Some visual comparison examples can be viewed in Fig. 6.

3) SynLiDAR → SemanticPOSS With XYZ as Input: As shown in Table III, our PMAN outperforms all compared methods by a large margin, which yields an accuracy of 46.5% in terms of mIoU. PMAN outperforms the source-only model by 8.2%. Compared to the second-best adversarial-based UDA segmentation method, Advent, we obtain considerable performance improvement (+6.4% in mIoU). Among all the 13 categories, PMAN achieves the best scores on eight categories and obtained comparable results in the rest five classes. Some visual comparison examples can be viewed in Fig. 7.

4) SynLiDAR → SemanticPOSS With XYZI as Input: As shown in Table IV, PMAN outperforms the source only model by 17.9%. Compared with PCT, PMAN is much simpler and offers the mIoU gain 7.8%. However, in comparison with SynLiDAR → SemanticKITTI, we obtain exactly the opposite experimental results. Taking the intensity as an input feature, the final performance of all methods drops. Compared with using *XYZ* as input, the performance of PMAN using XYZI as input feature drops the mIoU by 9.1%. This is because the intensity distribution of SynLiDAR and SmeanitcPOSS is dramatically different (as shown in Fig. 5). As stated above, each point intensity value of the SynLiDAR dataset is obtained from an intensity regression network trained on the SemanticKITTI dataset. Intensity will further increase the data distribution discrepancy between the source and target domains and result in a decline in the final performance, which suggests that directly using intensity as input is unwise. Nonetheless, our PMAN achieves the best performance in both cases, which proves its effectiveness.

5) Summary: First, since the intensity values of the SynLiDAR dataset are rendered by a model trained on the semanticKITTI dataset, it can improve the final performance on the SynLiDAR → SemanticKITTI task. However, the SemanticPOSS dataset is collected by a Pandora 40-line LiDAR which is different from the semanticKITTI (Velodyne HDL-64E LiDAR). The intensity will increase the domain gap on SynLiDAR → SemanticPOSS UDA segmentation task and degrade the final results. Thus, we should be careful to use intensity as an input feature in real-world UDA applications. Second, experimental results also show that there is still a large gap between UDA and fully supervised segmentation methods (i.e., Target only in Tables I–IV), and more deep research is needed in the future.

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TABLE IV

COMPARISON RESULTS OF SYNLIDAR → SEMANTICPOSS ADAPTATION IN TERMS OF MIOU. THE INPUT FEATURE OF ALL METHODS IS THE COMBINATION OF XYZ COORDINATES AND INTENSITY. "SOURCE ONLY" AND "TARGET ONLY" DENOTE THE MODEL ONLY TRAINED ON SOURCE DATA AND TARGET DATA WITHOUT ADAPTATION, RESPECTIVELY. THE MECHANISM "A" AND "S" DENOTE ADVERSARIAL TRAINING AND SELF-TRAINING, RESPECTIVELY

Methods	Mech.	bi.clst	car	trunk	veget.	traf.	pole	garb.	build.	cone.	fence	bi.cle	ground	pers.	mIoU
Source only	-	6.8	27.8	13.3	53.5	2.0	13.8	5.9	52.5	1.9	6.9	1.3	61.2	6.8	19.5
AdaptSegNet [11]	A	47.2	39.5	23.8	68.9	5.1	32.4	15.4	58.4	2.9	19.3	8.8	76.5	53.1	34.7
CLAN [14]	A	49.5	40.6	28.4	71.1	6.2	32.7	15.6	63.9	2.3	18.9	8.1	75.8	56.1	36.1
AdvEnt [13]	A	49.1	39.5	21.0	69.8	**7.0**	33.0	15.8	61.8	1.9	28.5	6.2	77.8	55.0	35.9
DAST [16]	A	45.9	38.2	22.4	69.3	3.7	23.1	13.8	61.5	3.1	14.3	7.6	73.7	48.9	32.7
FADA [12]	A	48.6	38.4	23.0	72.2	5.4	31.4	13.9	**64.5**	2.9	19.3	**9.4**	76.0	51.0	35.1
MRNet [49]	A	47.5	39.4	24.5	**79.0**	5.3	33.7	14.8	60.9	3.5	18.5	8.3	76.3	53.8	35.1
PCT [42]	S	34.8	27.8	18.6	63.7	4.9	**41.0**	16.6	64.1	1.6	12.1	6.6	63.9	28.9	29.6
PolarMix [51]	S	-	-	-	-	-	-	-	-	-	-	-	-	-	30.4
CoSMix [50]	S	31.6	**42.9**	**34.1**	59.5	0.5	23.0	17.3	52.3	1.8	1.1	2.3	**78.5**	16.6	27.8
PMAN (Ours)	A	**50.4**	39.1	28.4	71.3	4.3	35.2	**18.3**	63.3	**3.6**	**30.3**	5.6	76.8	**59.5**	**37.4**
Target only R [6]	-	48.9	68.1	41.7	75.4	28.5	31.4	11.4	73.8	27.7	44.9	53.9	82.1	51.5	49.2
Target only M [7]	-	63.0	71.2	45.4	77.7	48.6	42.7	27.8	75.7	45.4	59.2	57.2	80.9	70.8	58.9

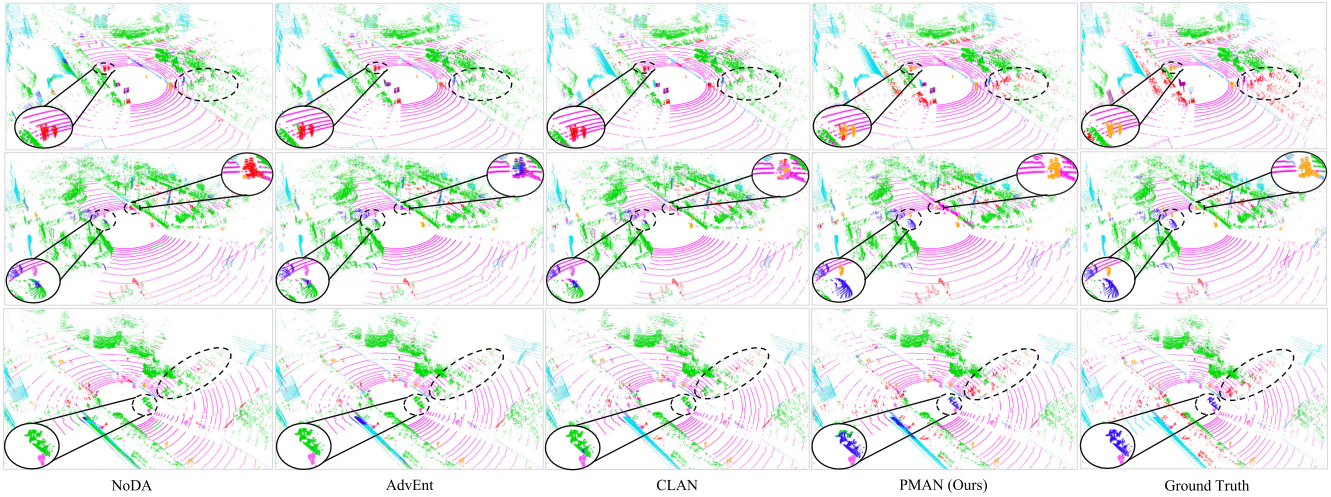

Fig. 7. Visual results of UDA segmentation for SynLiDAR → SemanticPOSS task. Black circles highlight some regions of interest.

TABLE V

ABLATION STUDY OF EACH PROPOSED COMPONENT OF PMAN WITH XYZ COORDINATES AND INTENSITY AS INPUT FEATURE

	IR^t	IR^s	PCAT	$\mathcal{L}^s_{ppd}$	$\mathcal{L}^t_{ppd}$	EntTh	PA	mIoU	gain
1								31.9	
2	✓							32.5	+0.6
3	✓	✓						31.8	-0.1
4			✓					32.6	+0.7
5	✓		✓					34.4	+2.5
6	✓		✓	✓				35.2	+3.3
7	✓		✓	✓	✓			31.5	-0.4
8	✓		✓	✓	✓	✓		35.8	+3.9
9	✓		✓		✓	✓	✓	32.4	+0.5
10	✓		✓	✓	✓	✓	✓	36.3	+4.4

V. ANALYSIS

A. Ablation Study

Table V reports the final segmentation performance of PMAN with certain components ablated. We use MinkUnet as the baseline method, the result of which is shown in the first row.

1) Intensity Regression: Rows 2–5 present the mIoU results of different intensity regression strategies. In particular, IR^t achieves +0.6% performance improvement in mIoU. As the intensity of SynLiDAR is predicted by a model which is different from the real data, when the intensity regression task is applied to the source data, the final performance drops by 0.1% (rows 1 and 3). With the combination of PCAT, the effectiveness of IR^t is confirmed by the significant performance gain, i.e., 1.8% in mIoU (rows 4 and 5). Furthermore, we utilize the IAS-Net without IR^t as the backbone network and set XYZ coordinates as the input feature to validate the effectiveness of the IR^t. As shown in Table VI, with the help of IR^t, we obtain +1.3% mIoU improvement on the final performance, which further proves the effectiveness of the IR^t.

2) Prototype-Guided Category-Level Adversarial Training: We further investigate the effectiveness of using the prototype to build an additional nonparametric classifier (i.e., C_a) to perform the category-level adversarial alignment. With the help of C_a, MinkUnet and IAS-Net obtain 0.7% (rows 1 and 4) and 1.9% (rows 2 and 5) improvements in mIoU accuracy, respectively. Beyond that, compared to CLAN which builds an

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TABLE VI

INFLUENCE OF IAS-NET INTENSITY REGRESSION BRANCH IN SYNLIDAR → SEMANTICKITTI UDA SEGMENTATION TASK. THE INPUT FEATURE IS XYZ COORDINATES

	with aux	w/o aux
mIoU	33.7	32.4

TABLE VII

EFFECT OF γ_1 IN THE TOTAL LOSS $\mathcal{L}_{\text{TOTAL}}$

γ_1	0.01	0.1	1	10
mIoU	36.3	36.3	36.0	33.3

additional parametric classifier to construct a different view, we do not require any extra learnable parameters and achieve similar performance gains.

3) $\mathcal{L}^s_{ppd}$ Loss: In Table V, comparing rows 5 and 6, by adding this loss, we get a 0.8% mIoU increase. And from rows 9 and 10, without $\mathcal{L}^s_{\text{ppd}}$ loss, the final model drops significantly (-3.9% in mIoU), confirming the benefit of this loss.

4) $\mathcal{L}^t_{ppd}$ Loss: In Table V, we conduct three ablated versions of $\mathcal{L}^t_{\text{ppd}}$ loss, including: 1) only using Th_p to get the pseudo-label ($\mathcal{L}^t_{\text{ppd}}$); 2) using both Th_p and Th_e to get the pseudo-label (EntTh); and 3) using the cos similarity to progressively minimize $\mathcal{L}^t_{\text{ppd}}$ (PA). As shown in rows 6 and 7, it is evident that only employing Th_p will result in many false pseudo-labels. As the error accumulates, the final performance drops 3.7%. The effectiveness of EntTh is confirmed by +0.6% mIoU improvement (rows 6 and 8). With the help of PA, the performance is further improved by 0.5% (rows 8 and 10), which proves the effectiveness of this strategy.

B. Parameter Sensitivity

1) γ_1: γ_1 is used to balance the auxiliary intensity regression task and other tasks. Therefore, we conduct experiments to observe the impact of changing this trade-off value. A larger γ_1 would make ISA-Net pay more attention to the intensity regression task. As our main task is UDA segmentation, giving too much emphasis on the auxiliary task may deteriorate the network discriminability. As shown in Table VII, when we set $\gamma_1 = 10$, the performance drops significantly, yielding an accuracy of 33.3% in mIoU. A proper choice of γ_1 is below 1.

2) γ_{local}: γ_{local} is a hyperparameter of the self-adaptive adversarial loss $\mathcal{L}^{t,\text{ada}}_{\text{adv}}$, which is utilized to enhance the influence of semantically consistent information during adversarial training. In this experiment, we fix $\epsilon = 0.4$ and experiment with different γ_{local} values. Table VIII presents the final segmentation performance of varying γ_{local}, among which our PMAN achieves the best result with $\gamma_{\text{local}} = 80$. When $\gamma_{\text{local}} = 20$, the $\mathcal{L}^{t,\text{ada}}_{\text{adv}}$ is dominated by the global adversarial loss, and the performance drops to 35.3% in mIoU. A proper choice of γ_{local} is between 60 and 100.

3) ϵ: For the self-adaptive loss $\mathcal{L}^{t,\text{ada}}_{\text{adv}}$, we aim to utilize the prediction consistency of C_a and C_m to obtain the easy- and hard-adapt points. ϵ is utilized to control the impact of the traditional global adversarial loss during the training process.

TABLE VIII

EFFECT OF γ_{LOCAL} IN THE SELF-ADAPTIVE ADVERSARIAL LOSS $\mathcal{L}^{t,\text{ADA}}_{\text{ADV}}$

γ_{local}	1	10	20	40	60	80	100
mIoU	29.6	34.6	35.3	35.8	36.2	36.3	36.3

TABLE IX

EFFECT OF ϵ IN THE SELF-ADAPTIVE ADVERSARIAL LOSS $\mathcal{L}^{t,\text{ADA}}_{\text{ADV}}$

ϵ	0.1	0.2	0.4	0.6	0.8
mIoU	36.3	36.3	36.3	36.2	36.2

TABLE X

EFFECT OF Th_e IN PSEUDO-LABEL SELECTION

Th_e	0.001	0.01	0.05	0.1
mIoU	35.7	36.3	35.4	33.4

TABLE XI

EFFECT OF Th_P IN PSEUDO-LABEL SELECTION

Th_P	0.9	0.93	0.95	0.97	0.99
mIoU	35.7	36.0	36.3	36.1	35.4

In Table IX, we fix $\gamma_{\text{local}} = 80$ and experiment with different ϵ values in the set of {0.1, 0.2, 0.4, 0.6, 0.8}. The experimental results show that the final performance is less sensitive to ϵ, and a proper choice is between 0.1 and 0.4.

4) Th_e: Th_e is utilized to choose the pseudo-label. A small Th_e would filter out many points with correct pseudo-labels that prevent PMAN from learning domain-invariant representations. A large Th_e would increase the risk of choosing the incorrect pseudo-label, which causes error accumulation and compromises the subsequent learning. The experimental results presented in Table X indicate that a small Th_e is better for the final result. A proper choice of Th_e is below 0.01.

5) Th_P: Th_p is also a hyperparameter which utilized to choose the pseudo-label. The larger the Th_p, the higher the confidence of the pseudo-label. In this experiment, we fix $\text{Th}_e = 0.01$ and experimental with several Th_p values in the set of {0.9, 0.93, 0.95, 0.97, 0.99}. Table XI shows the final results of varying Th_p values. Since we fixed Th_e into a small value, this hyperparameter is not sensitive to the final performance. However, a large Th_p (e.g., 0.99) will filter out many correct pseudo-labels and degrade the final performance. A proper choice of Th_p is between 0.93 and 0.97.

C. Effectiveness of the Auxiliary Classifier

As stated in Section III-B, the prototype-guided adversarial alignment is inspired by Cotraining [45] and CLAN [14]. Specifically, Cotraining utilizes two classifiers with different views on one sample to provide reliable pseudo-labels. When the segmentation model gives a target point a reliable prediction, the point is matched the source distribution to a large extent. One potential adverse effect is that these well-aligned points may be incorrectly mapped, leading to negative transfer in the subsequent training process [14]. Thus, we first utilize the prototype to construct a nonparametric auxiliary classifier C_a. Then, we use the prediction of the two

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TABLE XII

EXPERIMENTAL RESULTS OF C_a AND C_m ON SYNLIDAR → SEMANTICKITTI ADAPTATION IN TERMS OF MIOU. *XYZ* AND *I* DENOTE *XYZ* COORDINATES AND INTENSITY, RESPECTIVELY

Methods	Input	car	bi.cle	mt.cle	truck	oth-v.	pers.	bi.clst	mt.clst	road	parki.	sidew.	other-g.	build.	fence	veget.	trunk	terr.	pole	traf.	mIoU
Source Only C_a	XYZ	41.8	4.5	9.0	0.8	3.7	13.2	53.4	3.1	4.9	5.1	20.8	0.4	33.3	11.0	56.1	24.8	29.8	29.9	16.4	19.1
Source only C_m	XYZ	35.9	7.5	10.7	0.6	2.9	13.3	44.7	3.4	21.8	6.9	29.6	0.0	34.1	7.4	62.9	26.0	35.5	30.3	14.1	20.4
PMAN C_a	XYZ	70.7	7.8	23.1	1.6	6.3	23.5	62.7	5.6	73.7	9.1	54.5	0.2	47.3	16.9	69.5	37.7	61.9	37.2	19.3	33.1
PMAN C_m	XYZ	71.0	14.9	24.8	1.6	6.6	23.6	61.1	5.5	75.3	10.5	54.1	0.1	47.9	17.4	69.6	38.6	61.5	37.0	18.6	33.7
Source Only C_a	XYZI	44.1	6.9	21.4	0.5	4.0	15.8	20.9	0.5	5.6	5.1	23.1	0.3	34.9	16.1	60.0	31.8	37.7	28.0	15.8	19.6
Source only C_m	XYZI	42.5	8.6	14.6	0.6	3.8	15.3	49.0	1.3	19.4	5.5	28.0	0.1	37.8	10.3	63.1	31.5	41.6	28.1	11.3	21.7
PMAN C_a	XYZI	74.7	14.3	32.5	2.0	8.1	30.8	55.5	4.0	76.2	11.0	57.1	0.5	63.5	18.0	73.9	44.1	63.7	38.6	20.5	36.3
PMAN C_m	XYZI	74.9	18.0	33.3	1.9	7.8	28.7	52.7	4.0	77.6	12.6	56.9	0.1	63.8	17.9	74.1	43.8	63.5	38.3	19.3	36.3

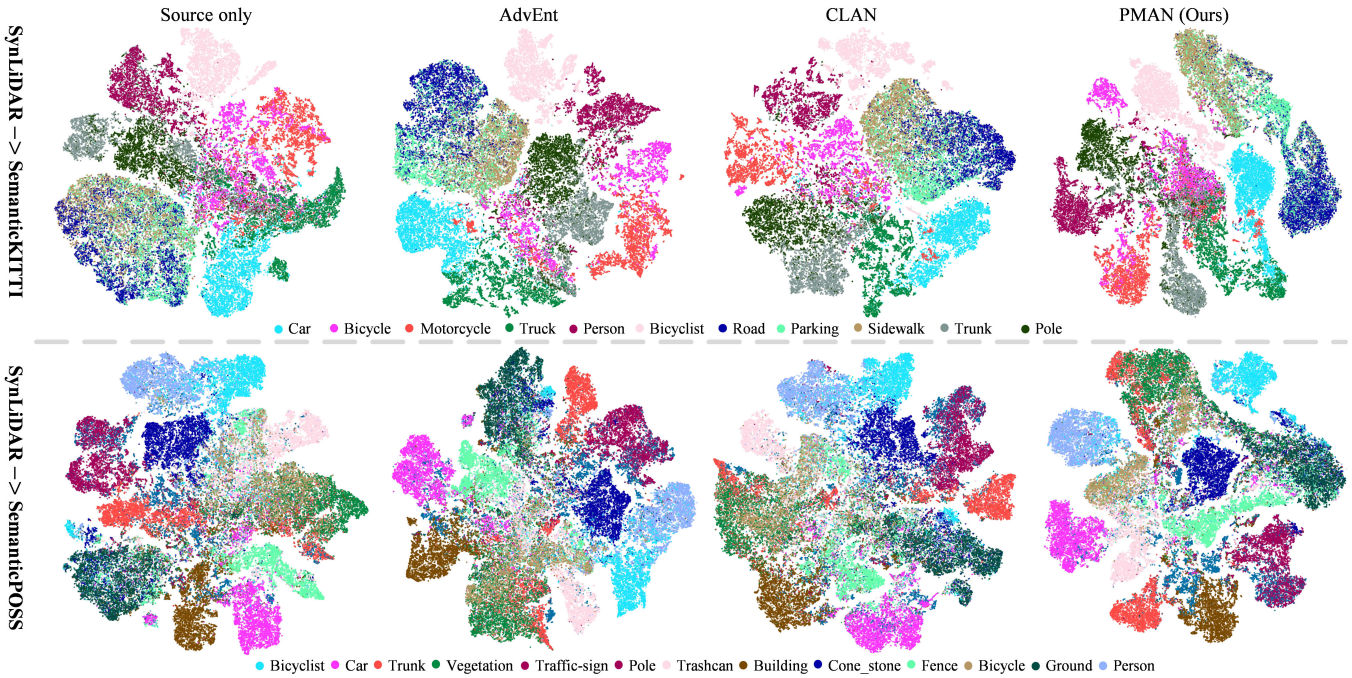

Fig. 8. t-SNE visualization of the embedded features on (Top) SynLiDAR → SemanticKITTI and (Bottom) SynLiDAR → SemanticPOSS UDA segmentation task. For a clear illustration, we only show 11 classes and different classes are denoted by different colors.

view classifiers (i.e., C_a and C_m) to generate the discrepancy map and adaptively reweighted the adversarial loss to solve this problem. As shown in Table XII, although the final mIoU scores of C_a and C_m are very close, each category IoU is very different. Thus, C_a and C_m can give two distinct predictions of a point and make prototype-guided adversarial alignment work in our PMAN.

D. Feature Visualization

In Fig. 8, we use t-SNE [52] to plot the embedded features of different methods on SynLiDAR → semanticKITTI and SynLiDAR → semanticPOSS UDA segmentation tasks. The multiple classes feature distributions of the source only model are mixed. Although AdvEnt and CLAN get a relatively compact feature distribution, the interclass distance between each class is pretty close. In comparison, the embedded features of our PMAN are better separated across distinct categories, which makes the subsequent classification easier.

VI. CONCLUSION

This article proposes a novel UDA framework, named PMAN, for urban scenes 3-D LiDAR point clouds semantic segmentation. The proposed PMAN constructs a multitask network by combining segmentation and intensity regression that can effectively learn generic features and perceive features of interest in the target domain. Meanwhile, an auxiliary nonparametric classifier is built using the source domain class prototype. We combine it with the main classifier to explicitly incorporate semantic consistency into the adversarial training process and effectively alleviate the side effects of traditional adversarial learning. Additionally, the prototype is applied to significantly reduce the domain shift at the category-level and

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shape the source domain to have better discriminative feature representation. Extensive experimental results demonstrate the superiority of PMAN on two benchmark datasets.

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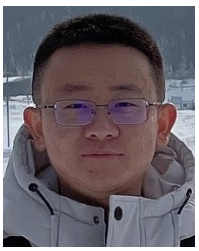

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